Foundation Alpha και Structured Tail Forecasting

> Ελληνική τεκμηρίωση strategy family. Το αρχείο είναι source-backed από τα YAML configs και τα διαθέσιμα logged artifacts στο repo. Δεν αποτελεί επενδυτική σύσταση και δεν υποκαθιστά ανεξάρτητο leakage/reproducibility review πριν από production χρήση.

Σύνοψη

ETHUSD 30m forecast-driven tail alpha με LightGBM/XGBoost/deep/foundation forecasters.

- Πλήθος checked-in configs που χαρτογραφήθηκαν: **25**.
- Πλήθος logged runs που χαρτογραφήθηκαν: **25**.
- Κύρια model kinds: `lightgbm_regressor` (16), `patchtst_forecaster` (2), `chronos_bolt_forecaster` (1), `lightgbm_clf` (1), `lstm_forecaster` (1), `sarimax_forecaster` (1), `tft_forecaster` (1), `timesfm_2p5_200m_forecaster` (1).
- Κύρια signal kinds: `forecast_threshold` (23), `forecast_threshold_hysteresis` (1), `meta_probability_side` (1).

Φιλοσοφία

Η οικογένεια `foundation_alpha` αντιμετωπίζει το trading ως πρόβλημα πρόβλεψης κατανομής forward return και όχι ως απλό indicator crossover. Η βασική θέση είναι ότι στα 30m ETHUSD υπάρχουν λίγες, ασύμμετρες ουρές απόδοσης που δεν πρέπει να γίνονται overtrade. Το strategy προσπαθεί να τις πιάσει μόνο όταν το forecast ξεπερνά αυστηρά thresholds.

Η αρχιτεκτονική είναι research-first: παραγωγή αιτιακών features, walk-forward εκπαίδευση, strict OOS predictions, μετατροπή forecast σε sparse signal και backtest με explicit κόστος. Τα configs v3.x είναι ablation surface γύρω από volatility/trend/cycle features, Ehlers replacements και GARCH variants.

Το σημαντικό design constraint είναι ότι το μοντέλο δεν πρέπει να βλέπει labels ή μελλοντικές τιμές. Τα lags, rolling windows, Ehlers features και target horizons πρέπει να παραμένουν χρονικά αιτιακά. Το καλύτερο run πρέπει να διαβάζεται ως OOS research result, όχι ως deployable claim.

Αρχιτεκτονική

- Data layer: cached Dukascopy 30m ETHUSD, UTC timestamps, PIT metadata και explicit processed snapshots.
- Feature layer: returns/volatility/trend/ATR, candlestick shape, Bollinger/RSI/StochRSI/MACD και Ehlers cycle/trend families στις v3.7 παραλλαγές.
- Model layer: κυρίως `lightgbm_regressor` με `future_return_regression`; συγκριτικά configs για XGBoost, LSTM, PatchTST, TFT, SARIMAX, Chronos και TimesFM.
- Signal layer: `forecast_threshold` ή συγγενικά threshold/hysteresis/vol filters που κάνουν sparse long/short exposure.
- Backtest/evaluation: strict OOS fold aggregation, fold diagnostics, feature importance, prediction coverage, cost drag και baseline diagnostics.

Causality, leakage και reproducibility guardrails

- Τα features πρέπει να υπολογίζονται μόνο από διαθέσιμες τιμές στο timestamp απόφασης. Rolling windows, lags και session aggregates δεν πρέπει να κοιτούν future bars.
- Τα model splits πρέπει να παραμένουν walk-forward/purged όπου ορίζεται, με OOS predictions να παράγονται μόνο για test rows.
- Τα labels, barrier outcomes και forward returns είναι training/evaluation targets. Δεν πρέπει να χρησιμοποιούνται στο signal layer εκτός OOS prediction output.
- Τα logged metrics πρέπει να διαβάζονται μαζί με costs, turnover, drawdown και fold dispersion. Υψηλό cumulative return με ασταθή folds ή τεράστιο cost drag δεν είναι robust edge.
- Τα links σε `logs/` είναι artifacts του τρέχοντος workspace. Αν λείπουν σε άλλο clone, το config παραμένει η canonical προδιαγραφή και το run πρέπει να αναπαραχθεί.

Inventory configs

Config	Strategy
BEST_ethusd_30m_lightgbm_h24_structured_tail_alpha_v3_7_ehlers_trend_hybrid.yaml	`ethusd_30m_lightgbm_h24_structured_tail_alpha_`
ethusd_30m_chronos_bolt_h24_tail_alpha_v1.yaml	`ethusd_30m_chronos_bolt_h24_tail_alpha_v1`
ethusd_30m_lightgbm_garch_overlay_h24_structured_tail_alpha_v3_7_ehlers_trend_hybrid.yaml	`ethusd_30m_lightgbm_garch_overlay_h24_structur`
ethusd_30m_lightgbm_garch_vol_filter_h24_structured_tail_alpha_v3_7_ehlers_trend_hybrid.yaml	`ethusd_30m_lightgbm_garch_vol_filter_h24_struc`
ethusd_30m_lightgbm_h12_structured_tail_alpha_v3_7_ehlers_trend_hybrid.yaml	`ethusd_30m_lightgbm_h12_structured_tail_alpha_`
ethusd_30m_lightgbm_h24_structured_tail_alpha_v1.yaml	`ethusd_30m_lightgbm_h24_structured_tail_alpha_`
ethusd_30m_lightgbm_h24_structured_tail_alpha_v2_hysteresis.yaml	`ethusd_30m_lightgbm_h24_structured_tail_alpha_`
ethusd_30m_lightgbm_h24_structured_tail_alpha_v2_no_raw_vol_levels.yaml	`ethusd_30m_lightgbm_h24_structured_tail_alpha_`
ethusd_30m_lightgbm_h24_structured_tail_alpha_v2_regime_gated.yaml	`ethusd_30m_lightgbm_h24_structured_tail_alpha_`
ethusd_30m_lightgbm_h24_structured_tail_alpha_v2_sparse_threshold_grid.yaml	`ethusd_30m_lightgbm_h24_structured_tail_alpha_`
ethusd_30m_lightgbm_h24_structured_tail_alpha_v3_1_remove_calm_abs.yaml	`ethusd_30m_lightgbm_h24_structured_tail_alpha_`
ethusd_30m_lightgbm_h24_structured_tail_alpha_v3_2_remove_calm_rank.yaml	`ethusd_30m_lightgbm_h24_structured_tail_alpha_`
ethusd_30m_lightgbm_h24_structured_tail_alpha_v3_3_remove_calm_bb_expansion.yaml	`ethusd_30m_lightgbm_h24_structured_tail_alpha_`

Config	Strategy
ethusd_30m_lightgbm_h24_structured_tail_alpha_v3_4_abs_range_bb_expansion.yaml	ethusd_30m_lightgbm_h24_structured_tail_alpha_
ethusd_30m_lightgbm_h24_structured_tail_alpha_v3_5_gk_additive.yaml	ethusd_30m_lightgbm_h24_structured_tail_alpha_
ethusd_30m_lightgbm_h24_structured_tail_alpha_v3_6_ehlers_trend_replacement.yaml	ethusd_30m_lightgbm_h24_structured_tail_alpha_
ethusd_30m_lightgbm_h24_structured_tail_alpha_v3_7_ehlers_trend_hybrid_tail_validation.yaml	ethusd_30m_lightgbm_h24_structured_tail_alpha_
ethusd_30m_lightgbm_h24_trade_quality_meta_v1.yaml	ethusd_30m_lightgbm_h24_trade_quality_meta_v1
ethusd_30m_lstm_h24_structured_tail_alpha_v3_7_ehlers_trend_hybrid.yaml	ethusd_30m_lstm_h24_structured_tail_alpha_v3_7_
ethusd_30m_patchtst_h24_structured_tail_alpha_v3_7_ehlers_trend_hybrid.yaml	ethusd_30m_patchtst_h24_structured_tail_alpha_
ethusd_30m_patchtst_h24_tail_alpha_baseline_v1.yaml	ethusd_30m_patchtst_h24_tail_alpha_baseline_v1
ethusd_30m_sarimax_h24_structured_tail_alpha_v3_7_ehlers_trend_hybrid.yaml	ethusd_30m_sarimax_h24_structured_tail_alpha_v
ethusd_30m_tft_h24_structured_tail_alpha_v3_7_ehlers_trend_hybrid.yaml	ethusd_30m_tft_h24_structured_tail_alpha_v3_7_
ethusd_30m_timesfm_2p5_h24_tail_alpha_v1.yaml	ethusd_30m_timesfm_2p5_h24_tail_alpha_v1
ethusd_30m_xgboost_h24_structured_tail_alpha_v3_7_ehlers_trend_hybrid.yaml	ethusd_30m_xgboost_h24_structured_tail_alpha_v

Best διαθέσιμο run

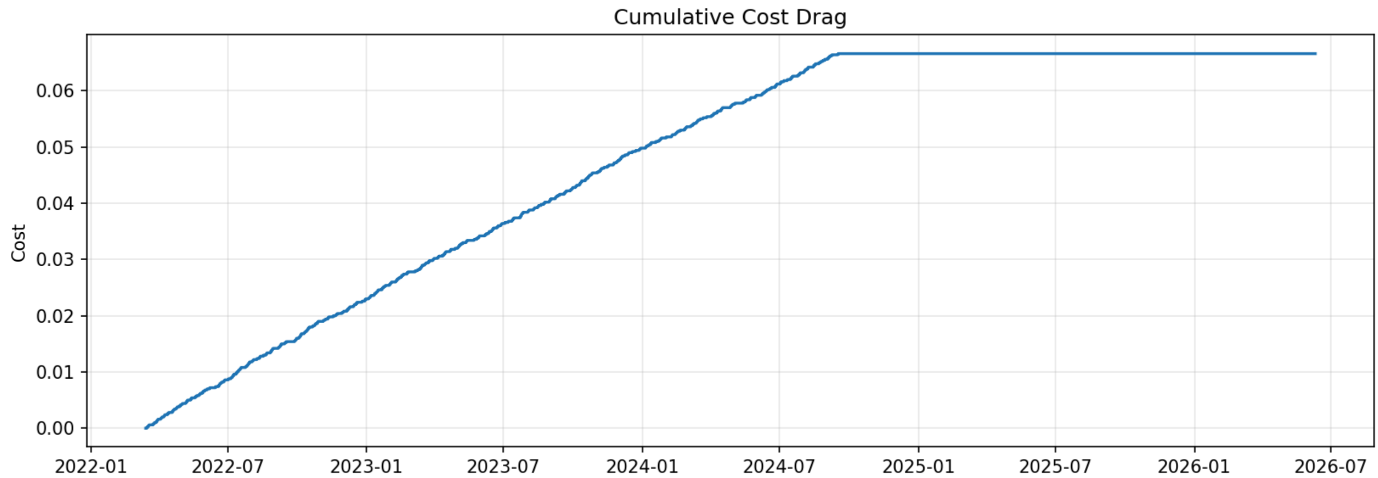
Το καλύτερο διαθέσιμο run επιλέχθηκε μηχανικά με προτεραιότητα στο Sharpe όπου υπάρχει, και μετά στο cumulative return/PnL. Αυτό δεν σημαίνει ότι είναι το πιο production-ready run· σημαίνει ότι είναι το ισχυρότερο logged αποτέλεσμα με το διαθέσιμο scoring rule.

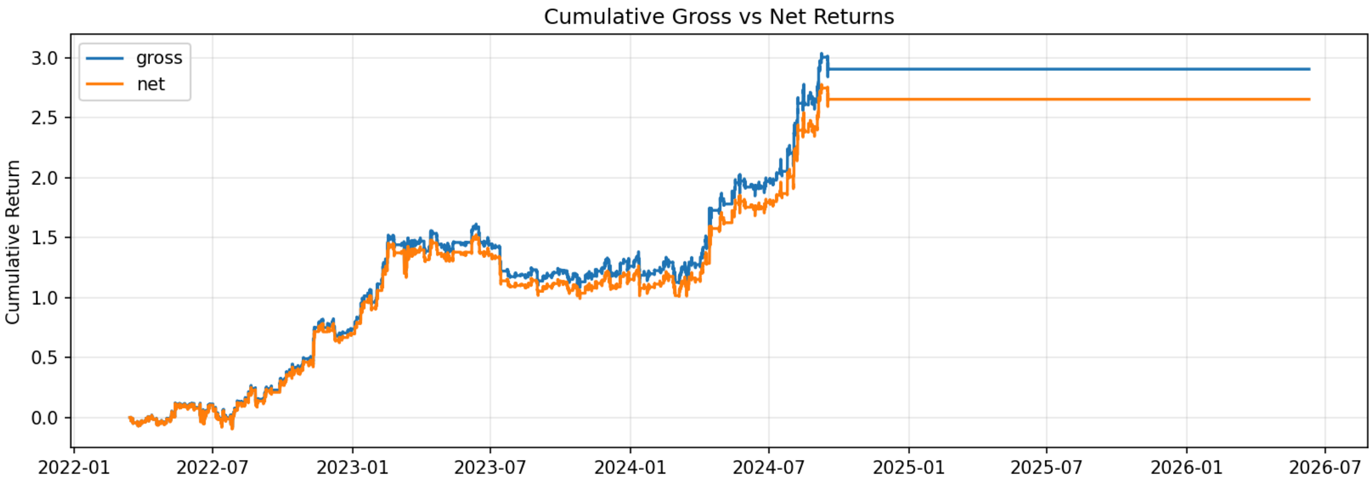
- Run: ethusd_30m_lightgbm_garch_overlay_h24_structured_tail_alpha_v3_7_ehlers_trend_hybrid_20260705_163128_876008_ae3471e3
- Config path από metadata: `config/experiments/foundation_alpha/ethusd_30m_lightgbm_garch_overlay_h24_structured_tail_alpha_v3_7_ehlers_trend_hybrid.yaml`
- Canonicality: **checked-in strategy config** στο `config/experiments/`.
- Strategy name: `ethusd_30m_lightgbm_garch_overlay_h24_structured_tail_alpha_v3_7_ehlers_trend_hybrid`
- Model / Signal: `lightgbm_regressor` / `forecast_threshold`
- Assets: ETHUSD

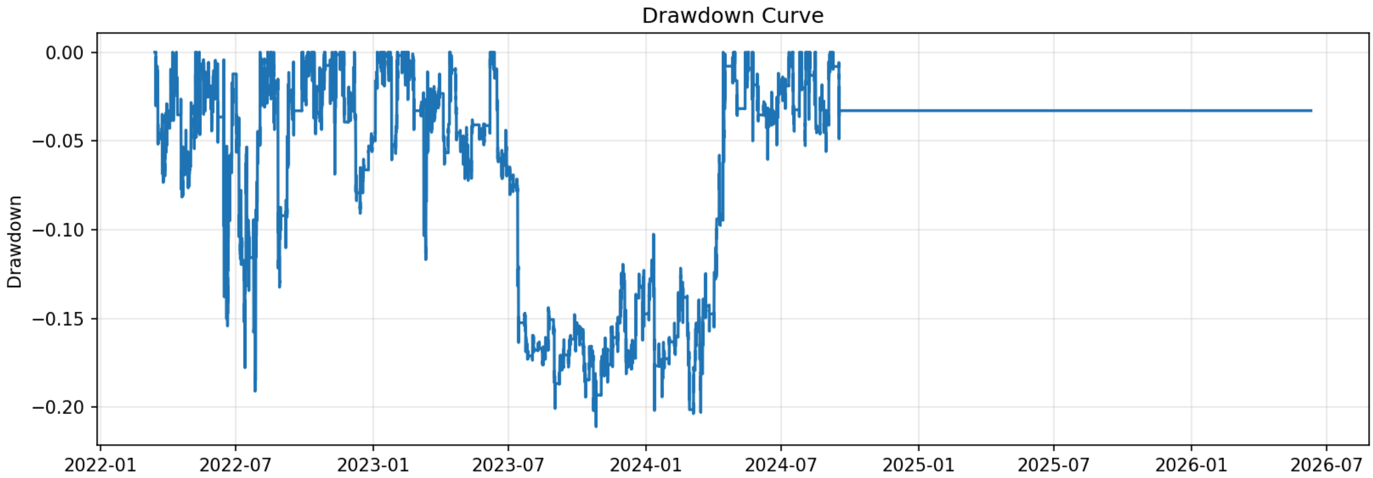
Metric	Τιμή
Cumulative return / total PnL	2.6539
Annualized return	0.6792
Sharpe	2.2953
Sortino	3.5375
Max drawdown	-0.2110
Profit factor	1.1142
Hit rate	0.4889
Total cost / fees	0.0666
Total turnover	666.0000

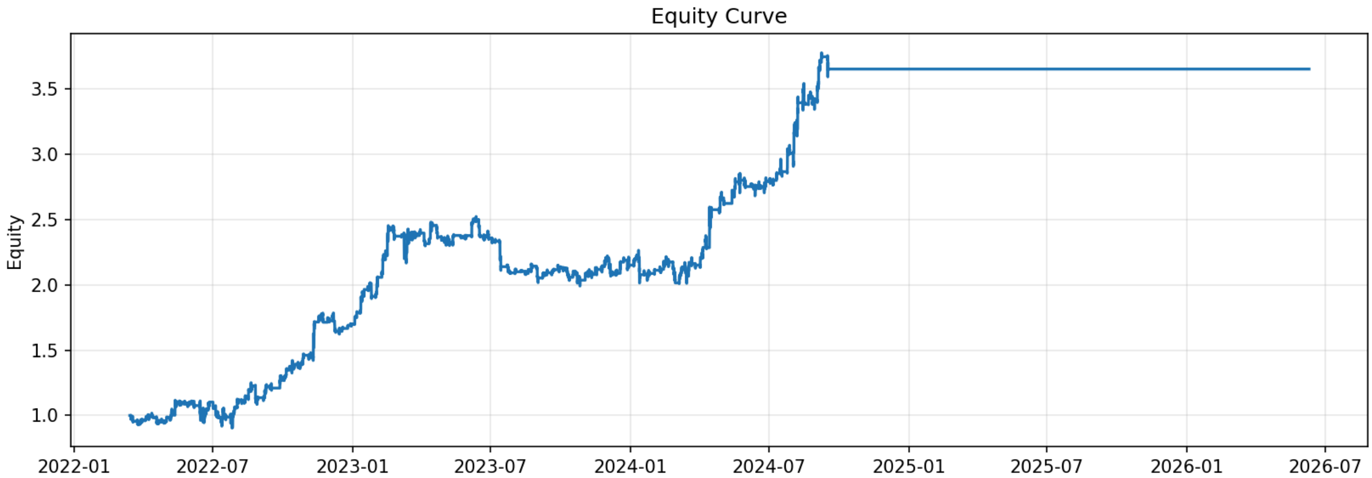
Κύριο report: report.md

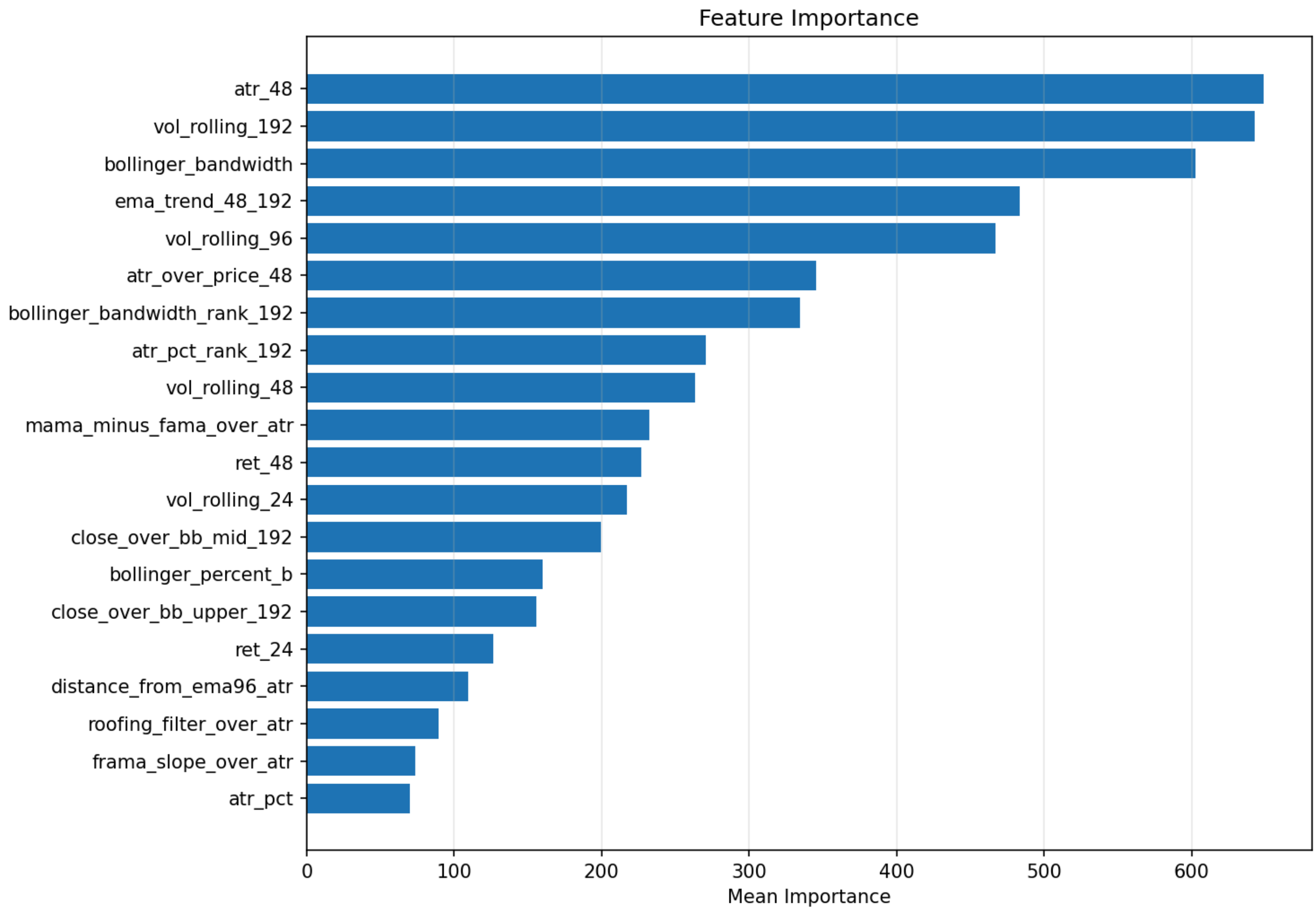
Plots του best run

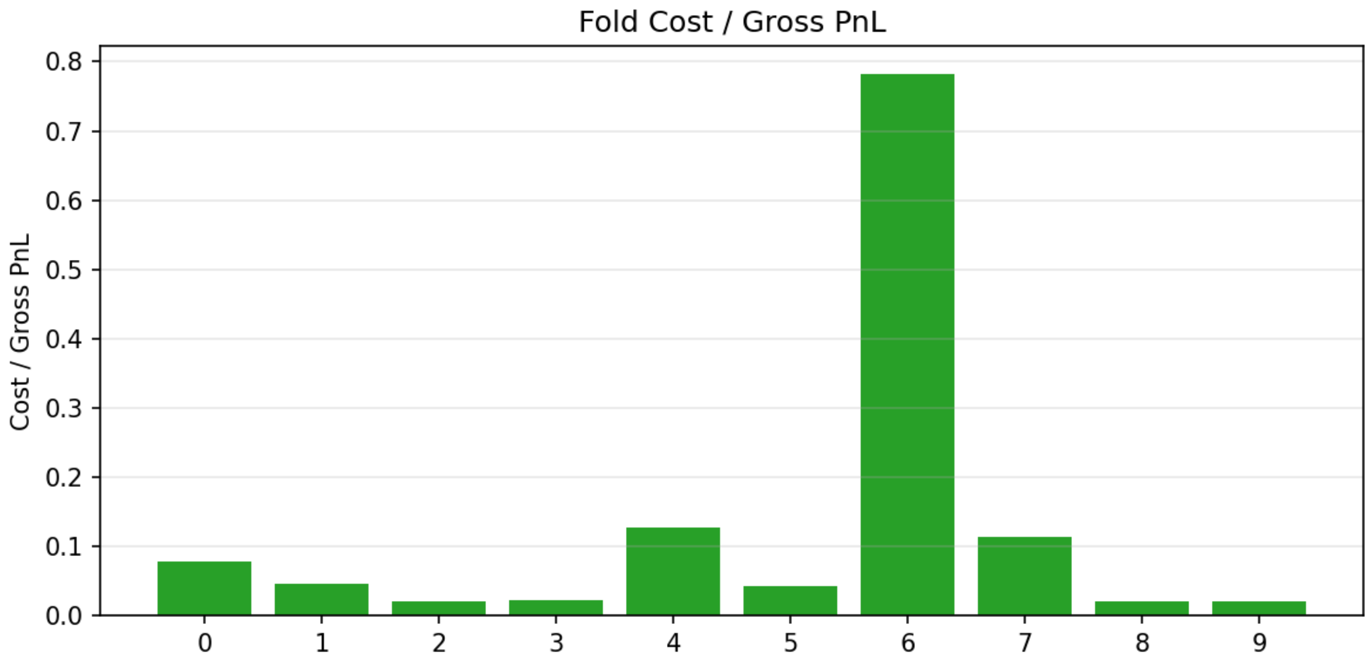


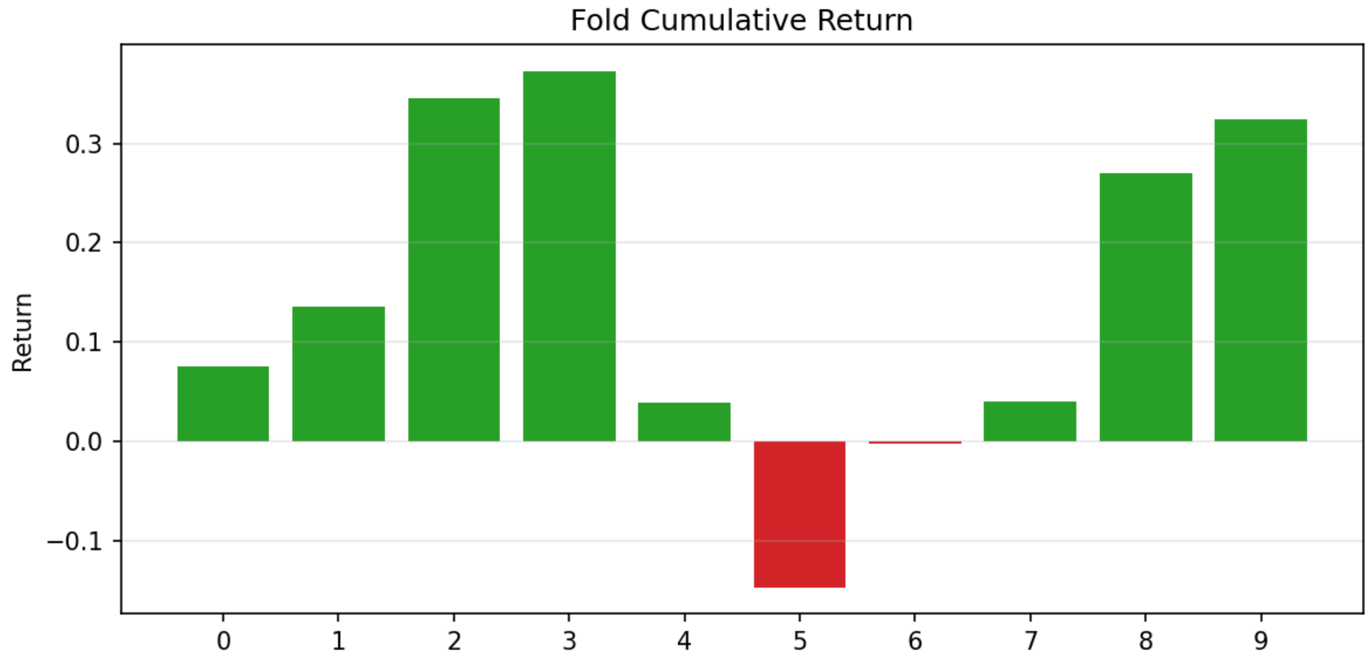

Cumulative Gross vs Net Returns
gross
net
Cumulative Return
3.0
2.5
2.0
1.5
1.0
0.5
0.0
2022-01
2022-07
2023-01
2023-07
2024-01
2024-07
2025-01
2025-07
2026-01
2026-07

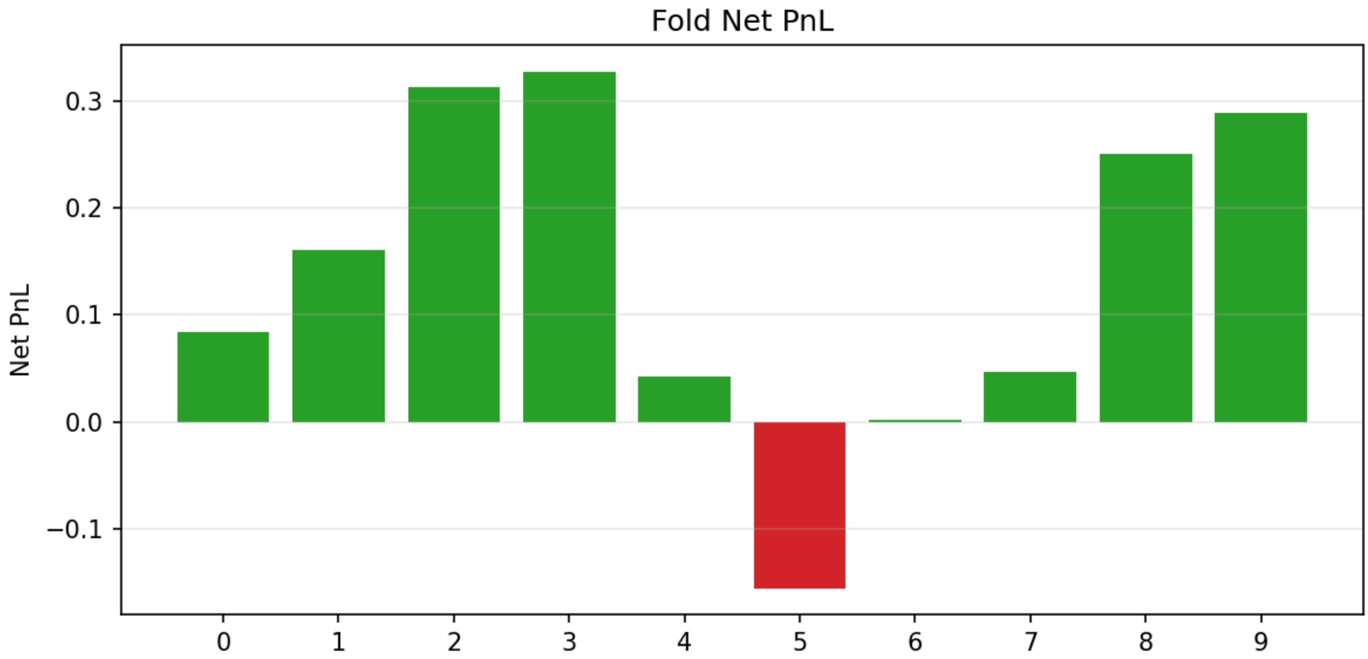

Drawdown Curve
Drawdown
0.00
−0.05
−0.10
−0.15
−0.20
2022-01
2022-07
2023-01
2023-07
2024-01
2024-07
2025-01
2025-07
2026-01
2026-07

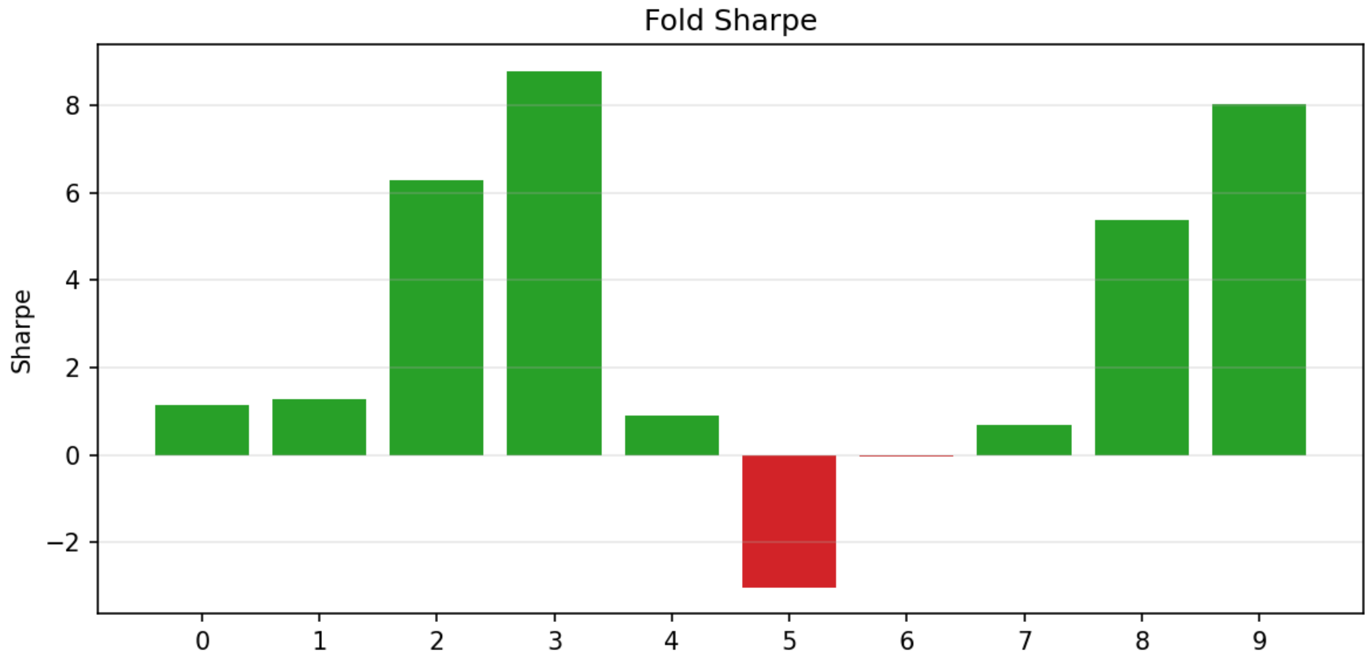

Equity Curve
Equity
3.5
3.0
2.5
2.0
1.5
1.0
2022-01
2022-07
2023-01
2023-07
2024-01
2024-07
2025-01
2025-07
2026-01
2026-07

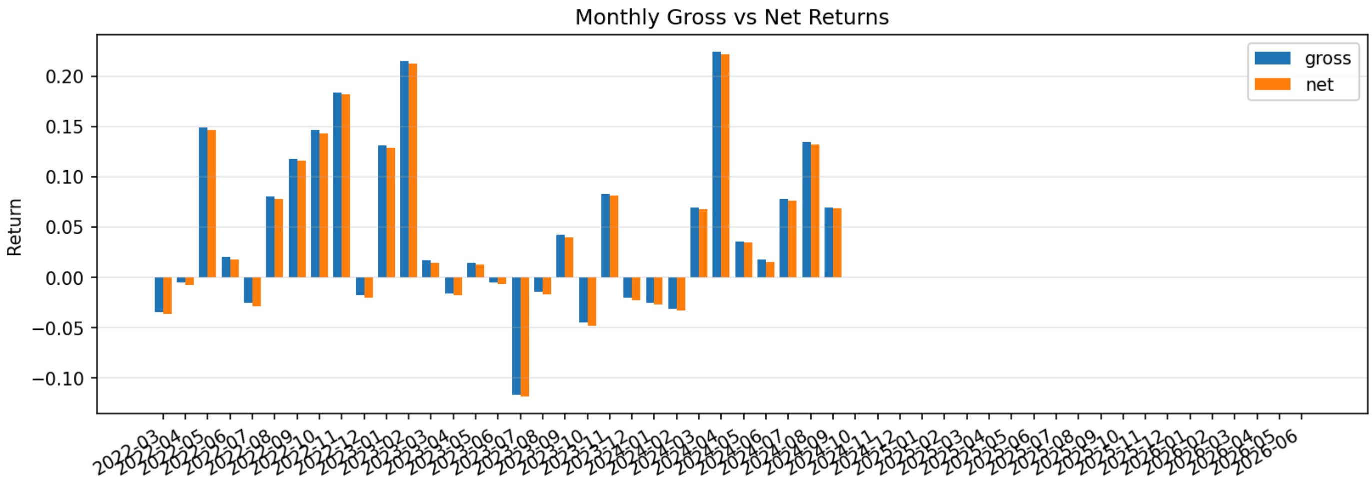

Feature Importance
atr_48
vol_rolling_192
bollinger_bandwidth
ema_trend_48_192
vol_rolling_96
atr_over_price_48
bollinger_bandwidth_rank_192
atr_pct_rank_192
vol_rolling_48
mama_minus_fama_over_atr
ret_48
vol_rolling_24
close_over_bb_mid_192
bollinger_percent_b
close_over_bb_upper_192
ret_24
distance_from_ema96_atr
roofing_filter_over_atr
frama_slope_over_atr
atr_pct
0
100
200
300
400
500
600
Mean Importance

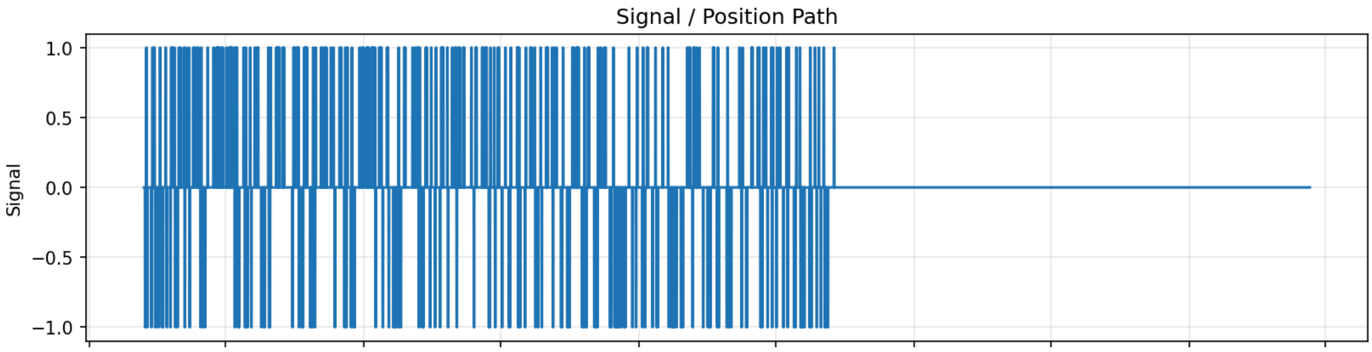

Fold Cost / Gross PnL
Cost / Gross PnL
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0
1
2
3
4
5
6
7
8
9

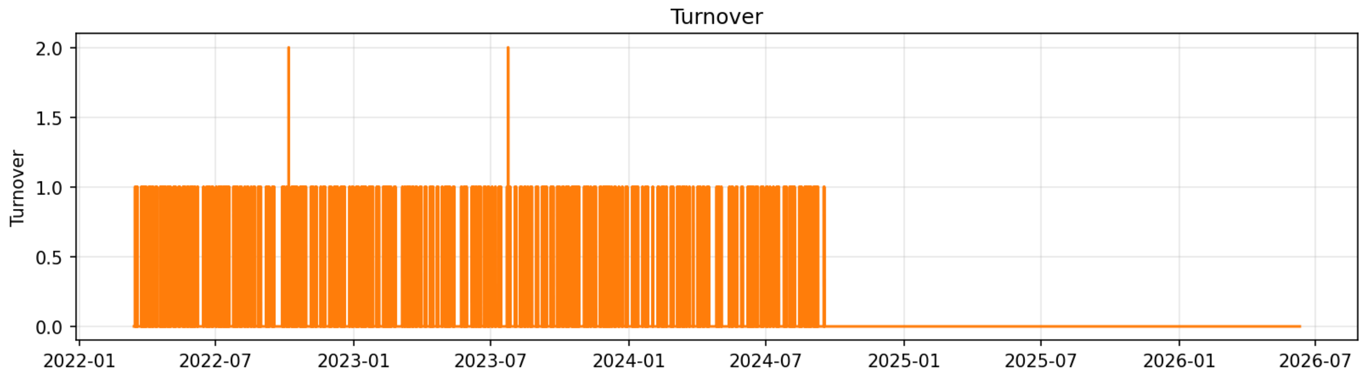

Fold Cumulative Return
Return
0.3
0.2
0.1
0.0
−0.1
0 1 2 3 4 5 6 7 8 9

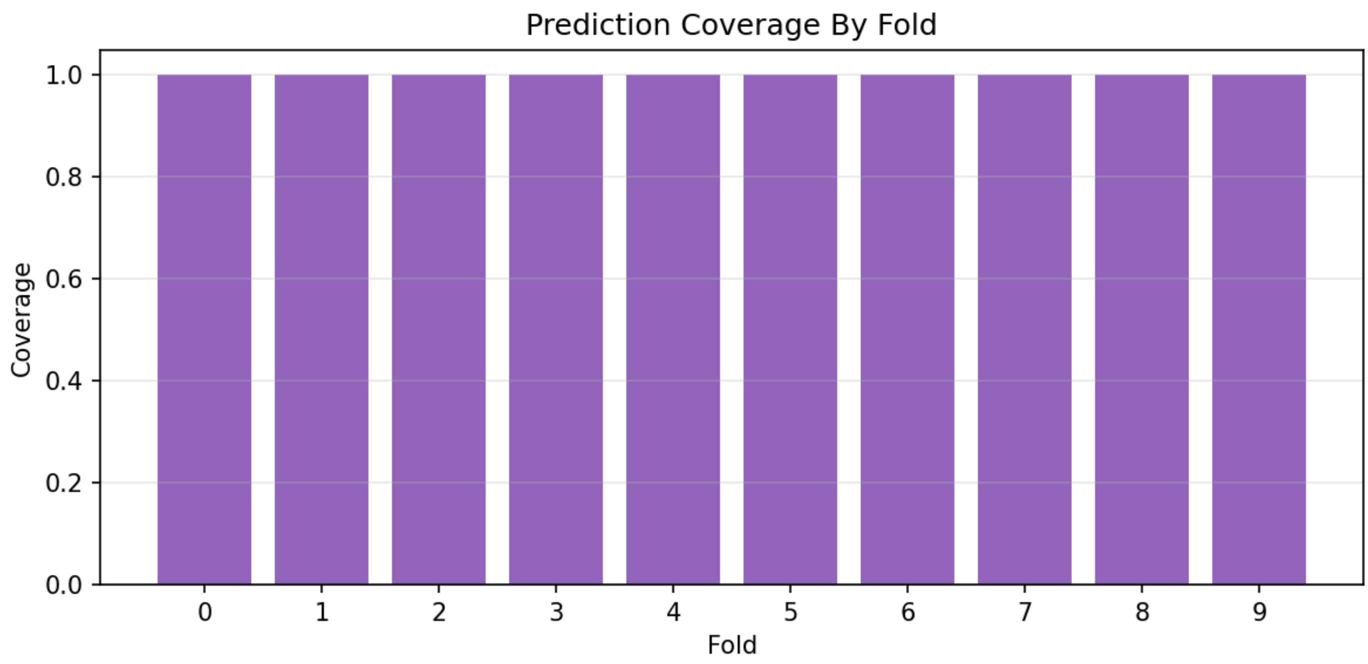

Fold Net PnL
Net PnL
0.3
0.2
0.1
0.0
−0.1
0 1 2 3 4 5 6 7 8 9

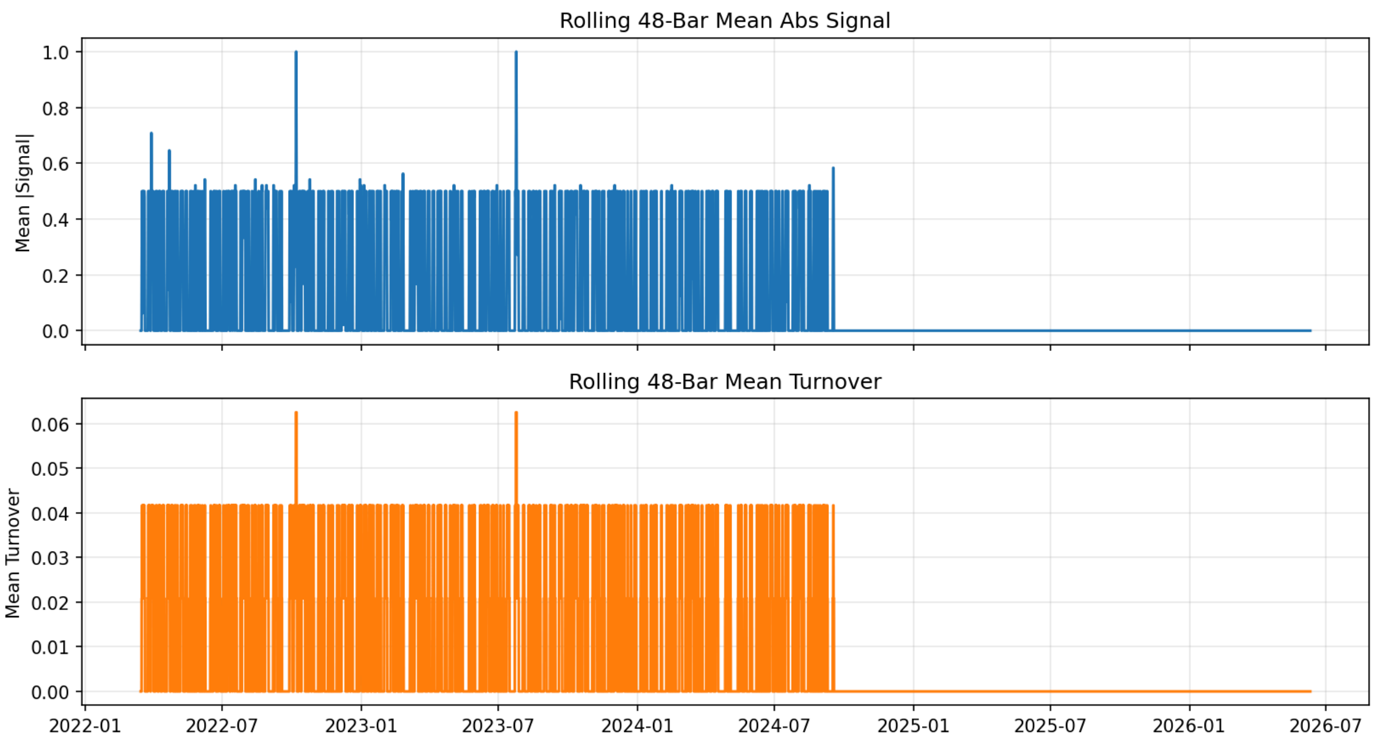

Fold Sharpe
Sharpe
8
6
4
2
0
−2
0 1 2 3 4 5 6 7 8 9

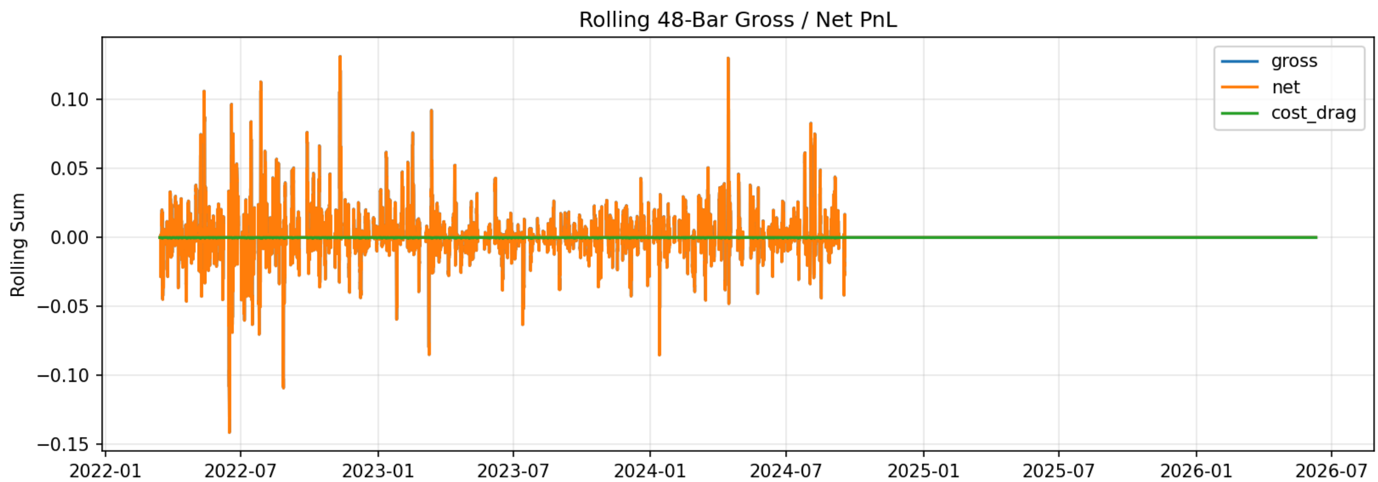

Monthly Gross vs Net Returns
Return
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
gross
net

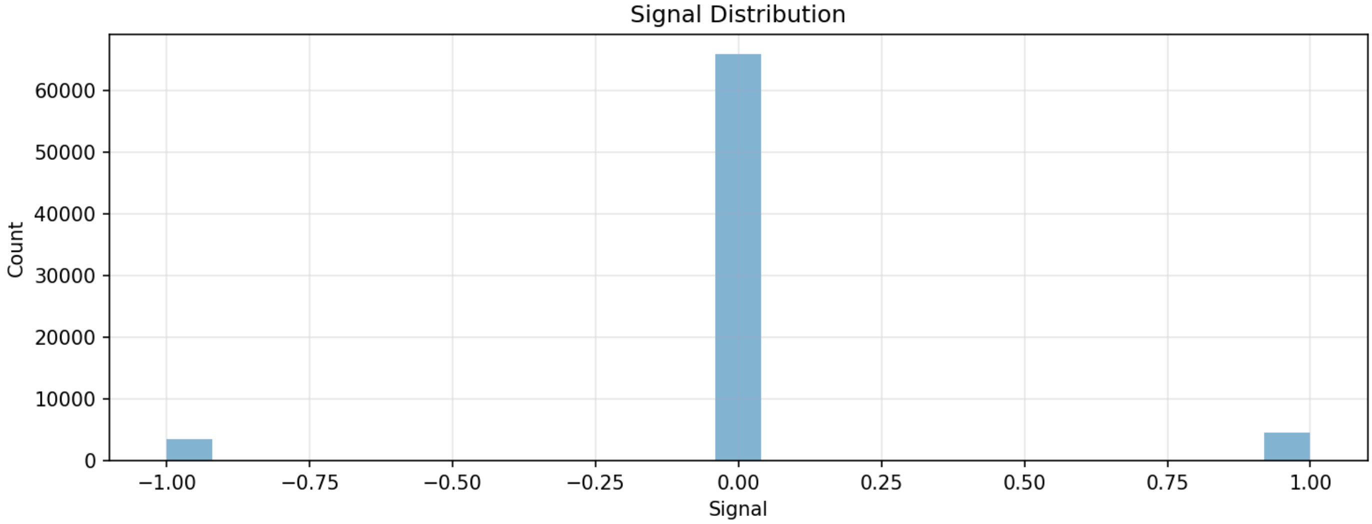

Signal / Position Path
Signal
1.0
0.5
0.0
−0.5
−1.0

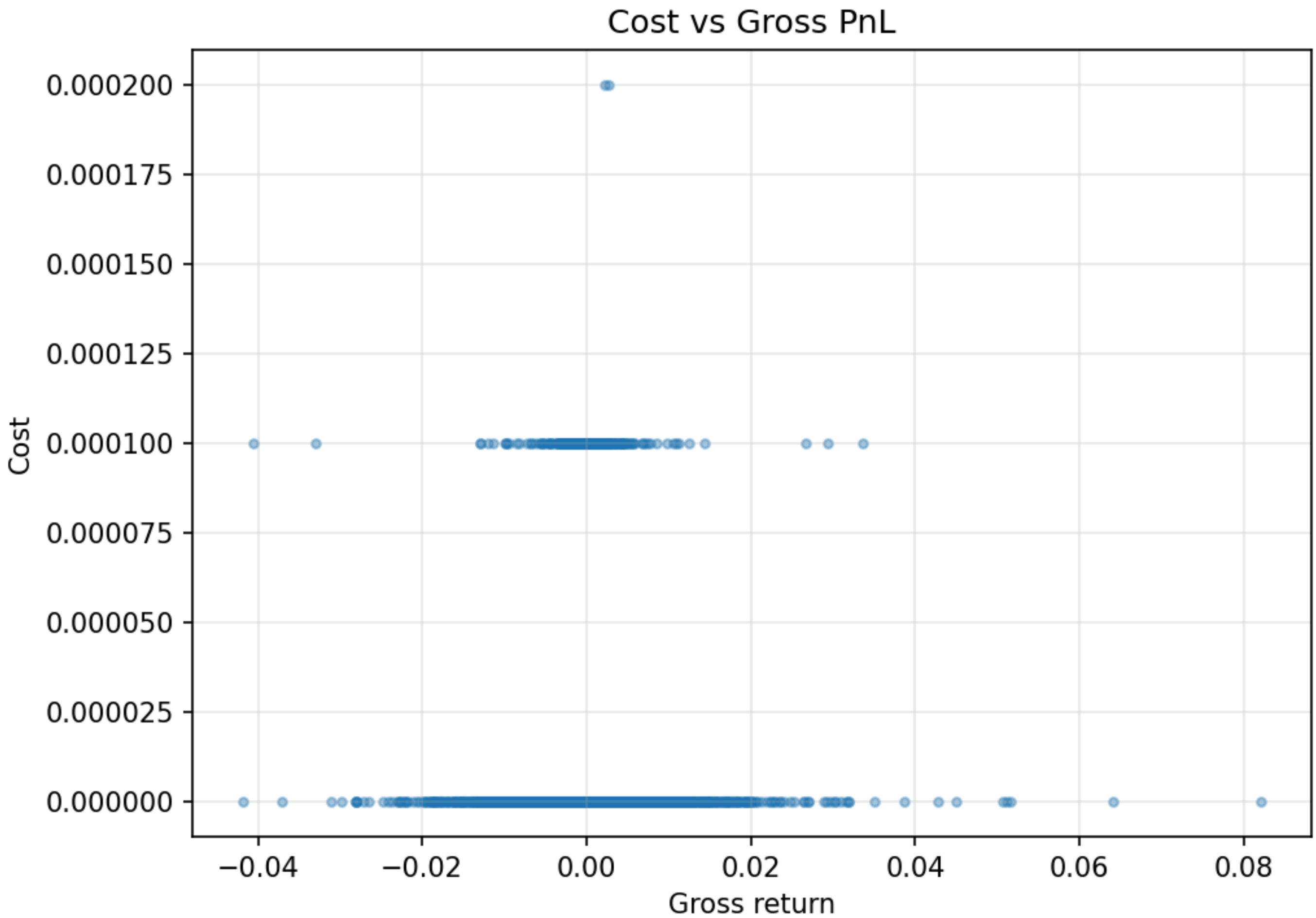

Turnover
Turnover
2.0
1.5
1.0
0.5
0.0
2022-01 2022-07 2023-01 2023-07 2024-01 2024-07 2025-01 2025-07 2026-01 2026-07

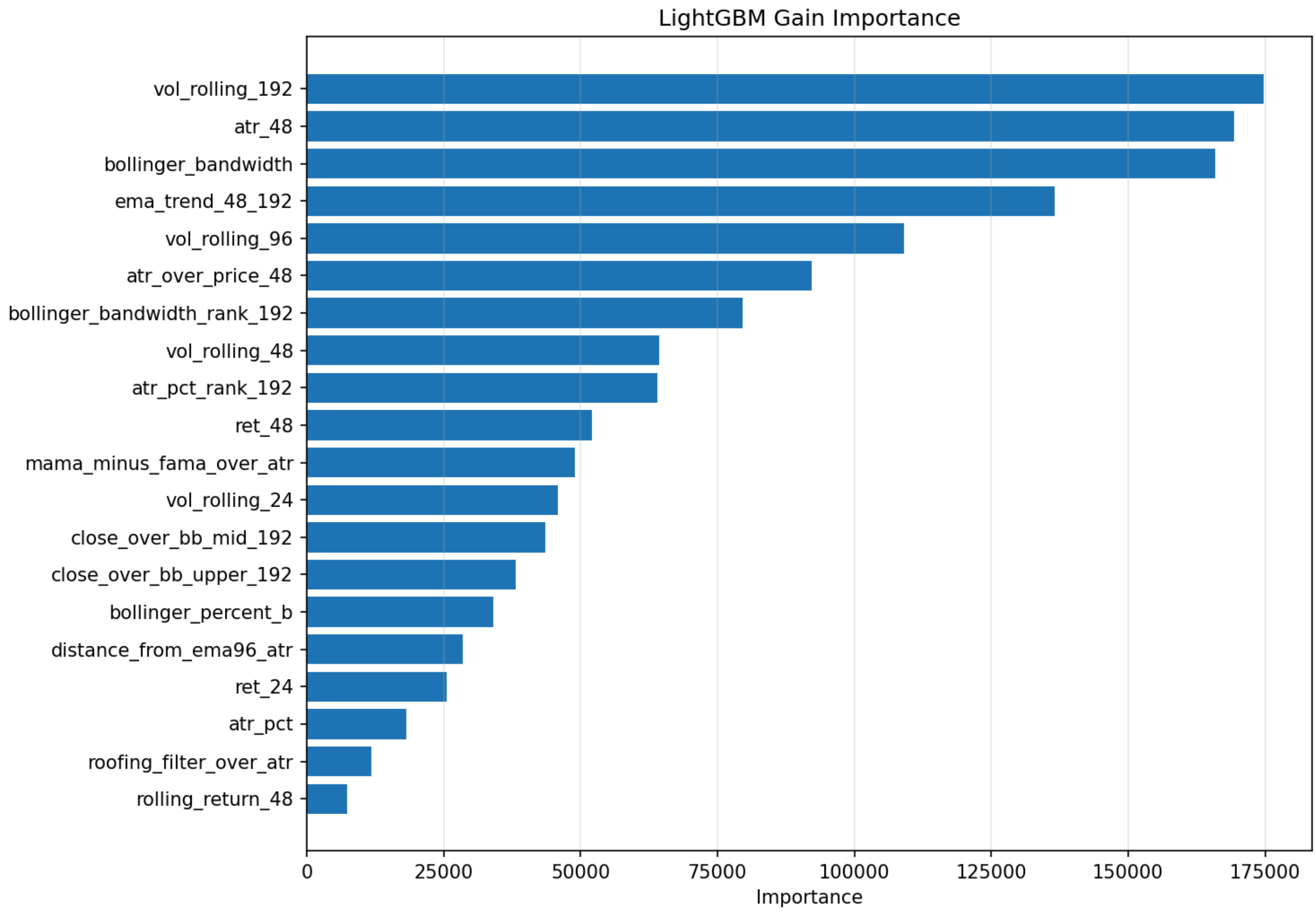

Prediction Coverage By Fold
Coverage
1.0
0.8
0.6
0.4
0.2
0.0
0 1 2 3 4 5 6 7 8 9
Fold

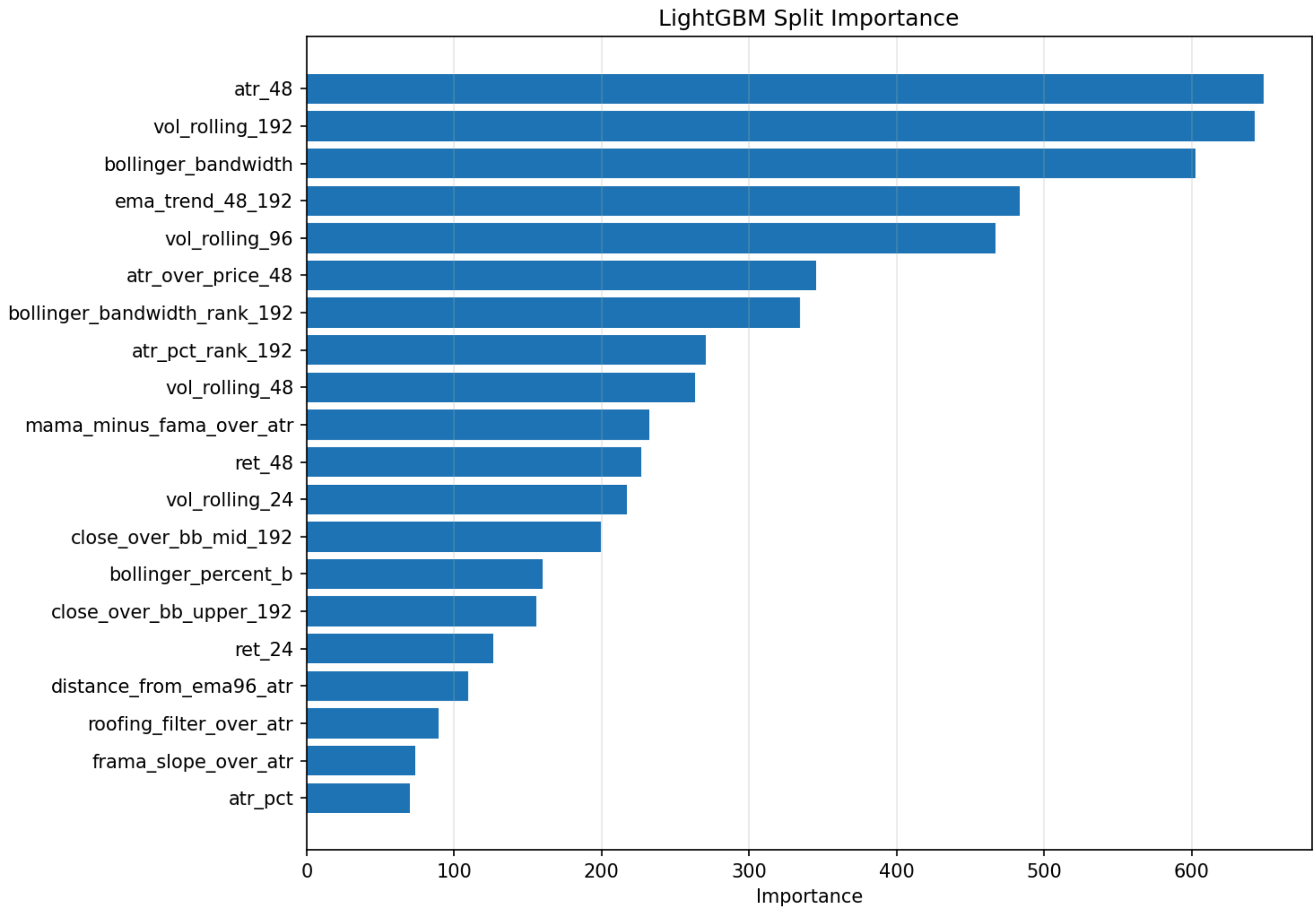

Rolling 48-Bar Mean Abs Signal
Mean |Signal|
1.0
0.8
0.6
0.4
0.2
0.0
Rolling 48-Bar Mean Turnover
Mean Turnover
0.06
0.05
0.04
0.03
0.02
0.01
0.00
2022-01 2022-07 2023-01 2023-07 2024-01 2024-07 2025-01 2025-07 2026-01 2026-07

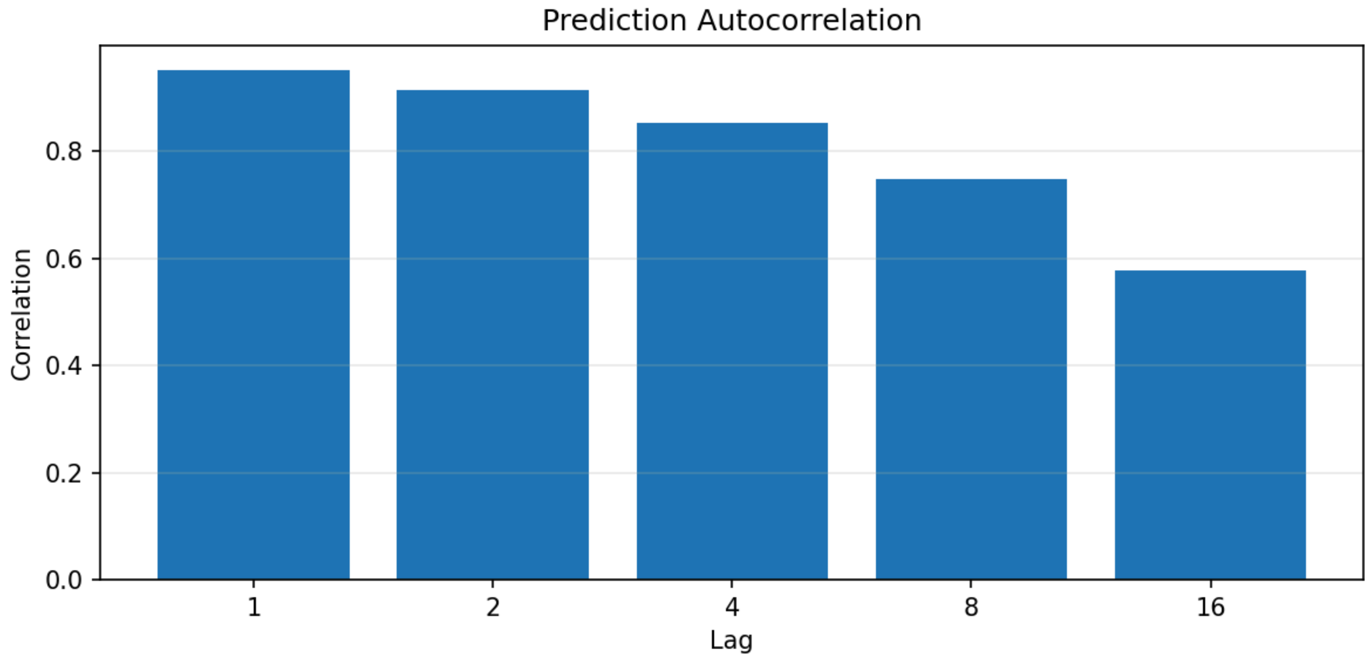

Rolling 48-Bar Gross / Net PnL
Rolling Sum
0.10
0.05
0.00
−0.05
−0.10
−0.15
gross
net
cost_drag
2022-01 2022-07 2023-01 2023-07 2024-01 2024-07 2025-01 2025-07 2026-01 2026-07

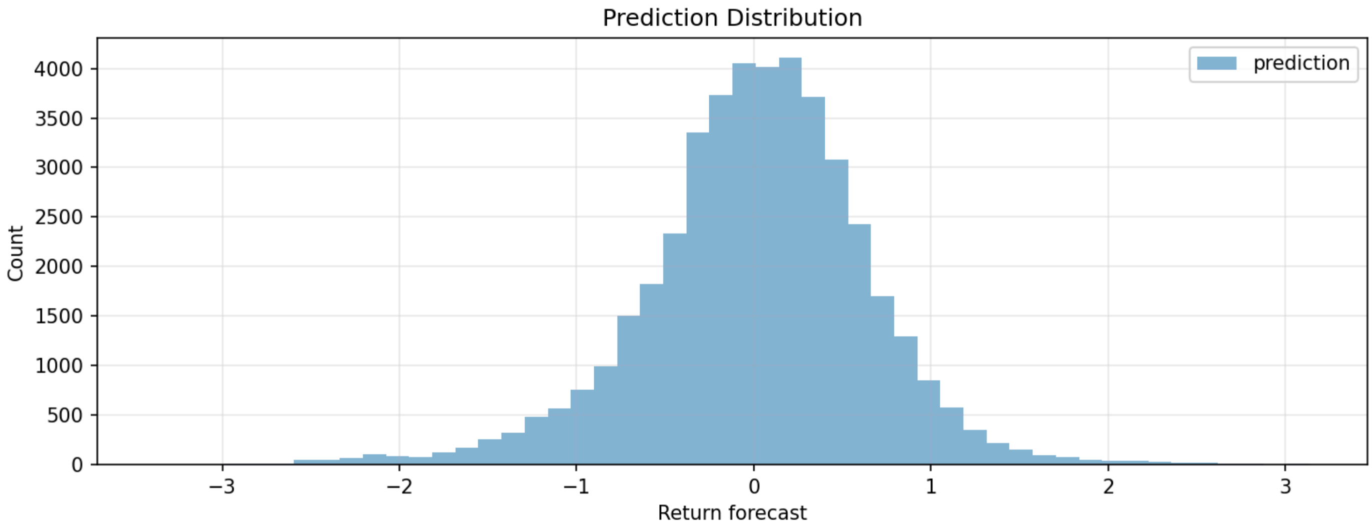

Signal Distribution
Count
Signal
60000
50000
40000
30000
20000
10000
0
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00

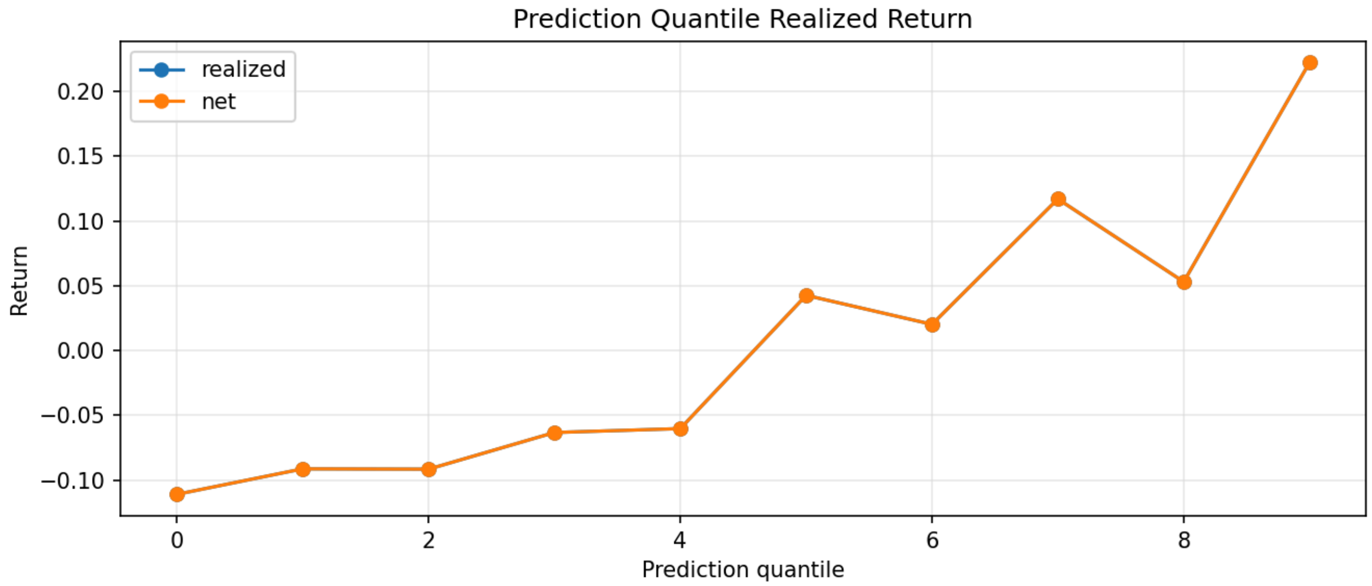

Cost vs Gross PnL
Cost
Gross return
0.000200
0.000175
0.000150
0.000125
0.000100
0.000075
0.000050
0.000025
0.000000
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08

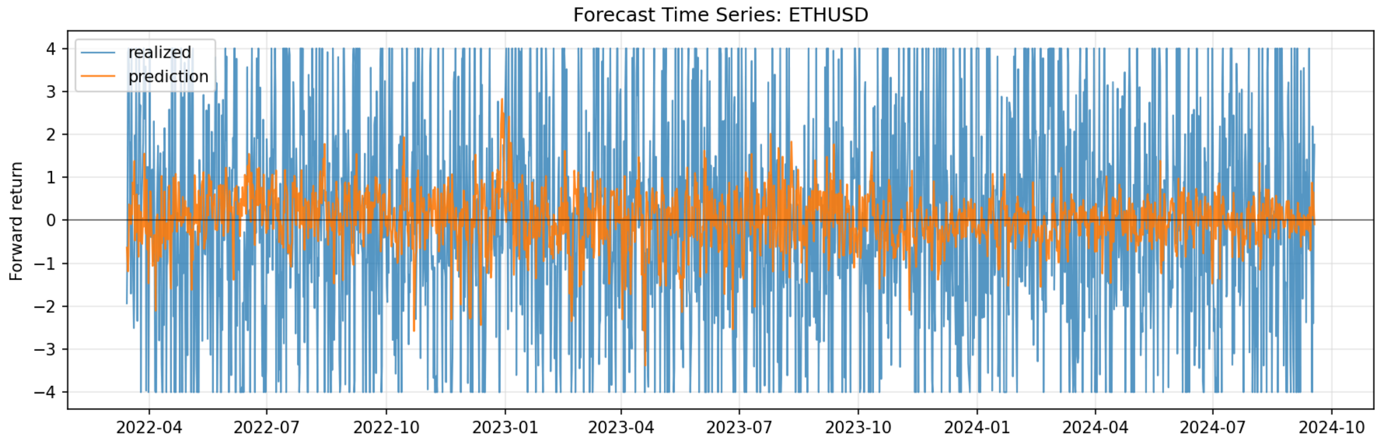

LightGBM Gain Importance
vol_rolling_192
atr_48
bollinger_bandwidth
ema_trend_48_192
vol_rolling_96
atr_over_price_48
bollinger_bandwidth_rank_192
vol_rolling_48
atr_pct_rank_192
ret_48
mama_minus_fama_over_atr
vol_rolling_24
close_over_bb_mid_192
close_over_bb_upper_192
bollinger_percent_b
distance_from_ema96_atr
ret_24
atr_pct
roofing_filter_over_atr
rolling_return_48
0
25000
50000
75000
100000
125000
150000
175000
Importance

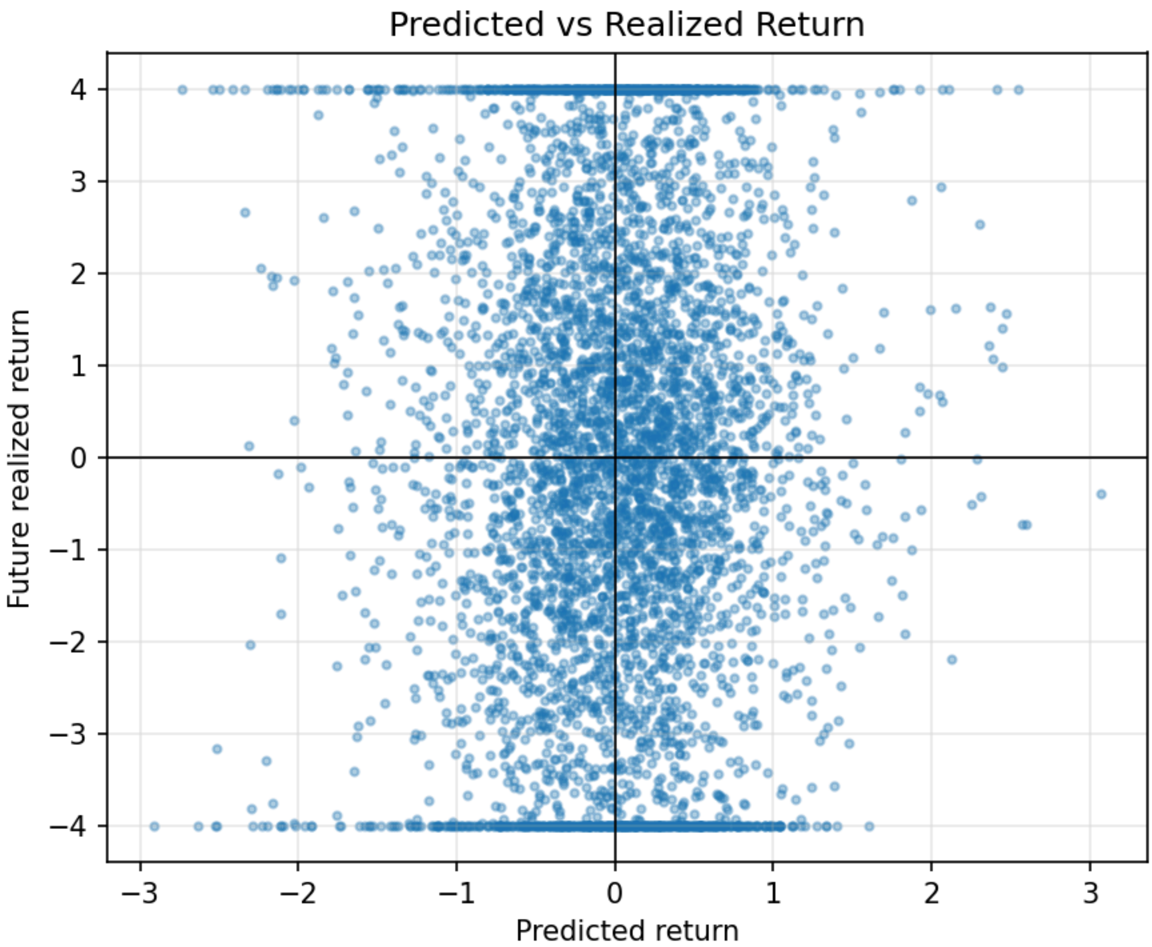

LightGBM Split Importance
atr_48
vol_rolling_192
bollinger_bandwidth
ema_trend_48_192
vol_rolling_96
atr_over_price_48
bollinger_bandwidth_rank_192
atr_pct_rank_192
vol_rolling_48
mama_minus_fama_over_atr
ret_48
vol_rolling_24
close_over_bb_mid_192
bollinger_percent_b
close_over_bb_upper_192
ret_24
distance_from_ema96_atr
roofing_filter_over_atr
frama_slope_over_atr
atr_pct
0
100
200
300
400
500
600
Importance

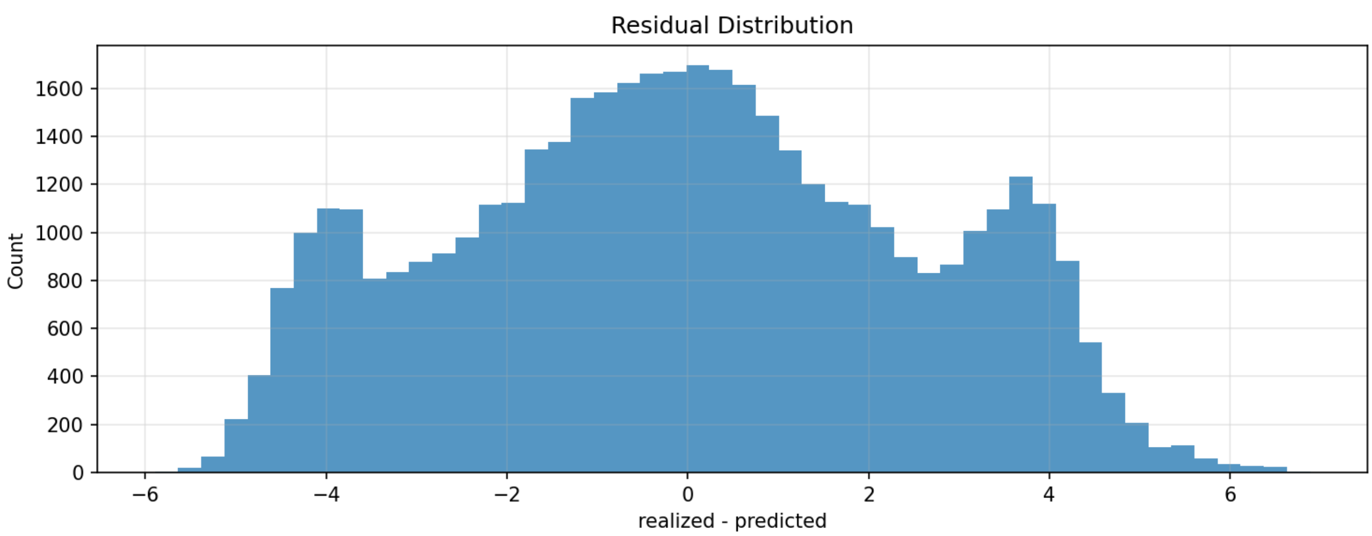

Prediction Autocorrelation
Correlation
0.0
0.2
0.4
0.6
0.8
1
2
4
8
16
Lag

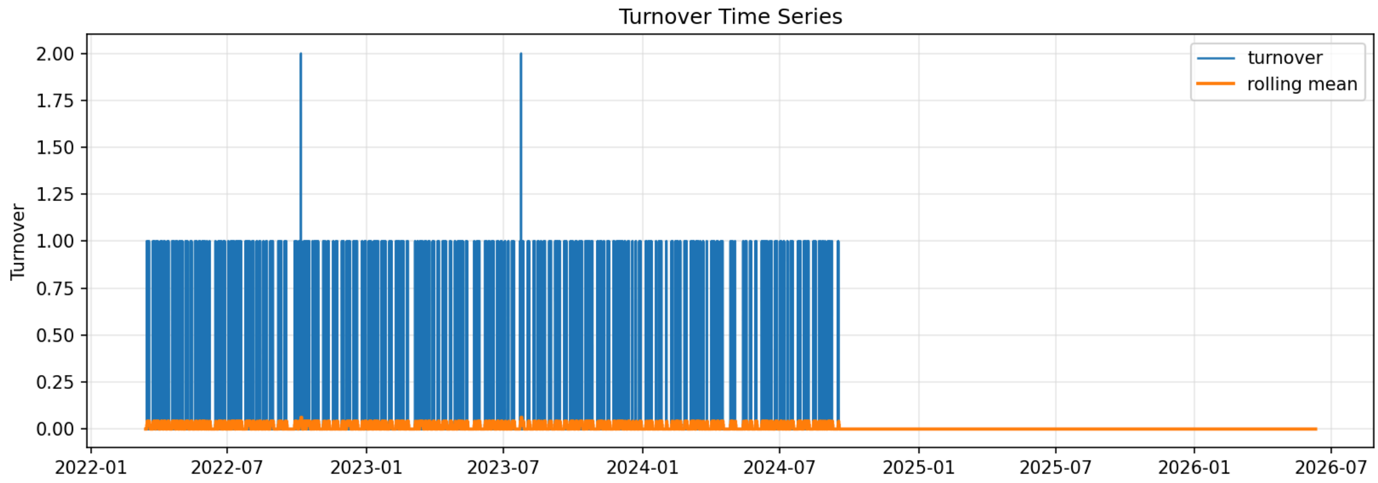

Prediction Distribution
prediction
Count
0
500
1000
1500
2000
2500
3000
3500
4000
−3
−2
−1
0
1
2
3
Return forecast

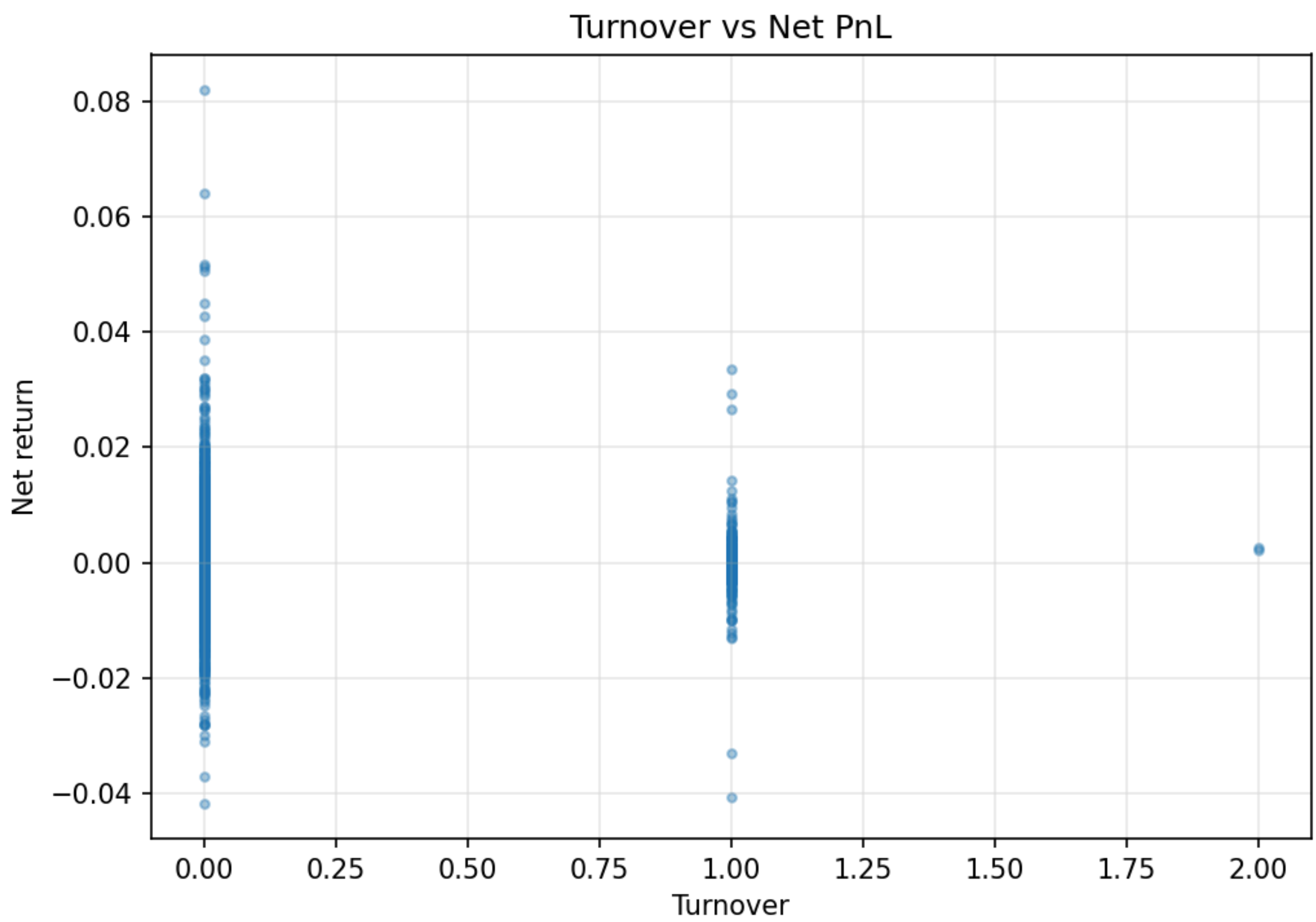

Prediction Quantile Realized Return
realized
net
Return
−0.10
−0.05
0.00
0.05
0.10
0.15
0.20
0
2
4
6
8
Prediction quantile


Forecast Time Series: ETHUSD
realized
prediction
Forward return
4
3
2
1
0
−1
−2
−3
−4
2022-04
2022-07
2022-10
2023-01
2023-04
2023-07
2023-10
2024-01
2024-04
2024-07
2024-10


Predicted vs Realized Return
Future realized return
Predicted return
4
3
2
1
0
−1
−2
−3
−4
−3
−2
−1
0
1
2
3


Residual Distribution
Count
realized - predicted
1600
1400
1200
1000
800
600
400
200
0
−6
−4
−2
0
2
4
6





Πλήρες artifact inventory του best run

Artifact	Ρόλος
report.md	Κύριο Markdown report του runner.
summary.json	Μηχανικά αναγνώσιμη σύνοψη metrics.
run_metadata.json	Metadata αναπαραγωγής, path config και runtime info.
config_used.yaml	Το ακριβές resolved config που χρησιμοποιήθηκε στο run.
prediction_diagnostics.json	Forecast/model diagnostic payload.
monitoring_report.json	Drift/monitoring diagnostics.
fold_model_summary.csv	CSV artifact για audit ή περαιτέρω ανάλυση.
feature_importance.csv	CSV artifact για audit ή περαιτέρω ανάλυση.
returns.csv	CSV artifact για audit ή περαιτέρω ανάλυση.

Artifact	Ρόλος
gross_returns.csv	CSV artifact για audit ή περαιτέρω ανάλυση.
equity_curve.csv	CSV artifact για audit ή περαιτέρω ανάλυση.
positions.csv	CSV artifact για audit ή περαιτέρω ανάλυση.
costs.csv	CSV artifact για audit ή περαιτέρω ανάλυση.
turnover.csv	CSV artifact για audit ή περαιτέρω ανάλυση.
artifact_manifest.json	JSON diagnostic/metadata artifact.
report_assets/cumulative_cost_drag.png	Σωρευτική επίδραση κόστους.
report_assets/cumulative_returns.png	Σωρευτικές αποδόσεις.
report_assets/drawdown_curve.png	Οπτική απεικόνιση drawdown.
report_assets/equity_curve.png	Οπτική απεικόνιση equity curve.
report_assets/feature_importance.png	Feature importance plot.
report_assets/fold_cost_to_gross_pnl.png	Plot artifact από το report/diagnostics.
report_assets/fold_cumulative_return.png	Plot artifact από το report/diagnostics.
report_assets/fold_net_pnl.png	Net PnL ανά fold.
report_assets/fold_sharpe.png	Sharpe ανά fold.
report_assets/monthly_returns.png	Μηνιαίες αποδόσεις.
report_assets/positions_turnover.png	Θέσεις και turnover.
report_assets/prediction_coverage_by_fold.png	Κάλυψη προβλέψεων ανά fold.
report_assets/rolling_behavior.png	Rolling behavior diagnostics.
report_assets/rolling_pnl.png	Rolling PnL.
report_assets/signal_distribution.png	Plot artifact από το report/diagnostics.
report_assets/trade_events.csv	Trade events για audit.
artifacts/diagnostics/cost_vs_gross_pnl.png	Plot artifact από το report/diagnostics.
artifacts/diagnostics/fold_backtest_diagnostics.csv	CSV artifact για audit ή περαιτέρω ανάλυση.
artifacts/diagnostics/forecast_alpha_diagnostics_summary.json	JSON diagnostic/metadata artifact.
artifacts/diagnostics/forecast_baselines.csv	CSV artifact για audit ή περαιτέρω ανάλυση.
artifacts/diagnostics/lgbm_gain_importance.png	Plot artifact από το report/diagnostics.
artifacts/diagnostics/lgbm_split_importance.png	Plot artifact από το report/diagnostics.
artifacts/diagnostics/lightgbm_importance.csv	CSV artifact για audit ή περαιτέρω ανάλυση.
artifacts/diagnostics/prediction_autocorrelation.csv	CSV artifact για audit ή περαιτέρω ανάλυση.
artifacts/diagnostics/prediction_autocorrelation.png	Plot artifact από το report/diagnostics.
artifacts/diagnostics/prediction_distribution.csv	CSV artifact για audit ή περαιτέρω ανάλυση.
artifacts/diagnostics/prediction_histogram.png	Plot artifact από το report/diagnostics.
artifacts/diagnostics/prediction_metrics.csv	CSV artifact για audit ή περαιτέρω ανάλυση.
artifacts/diagnostics/prediction_quantiles.csv	CSV artifact για audit ή περαιτέρω ανάλυση.
artifacts/diagnostics/prediction_quantiles.png	Plot artifact από το report/diagnostics.
artifacts/diagnostics/prediction_timeseries.png	Plot artifact από το report/diagnostics.
artifacts/diagnostics/prediction_vs_realized.png	Plot artifact από το report/diagnostics.
artifacts/diagnostics/regime_diagnostics.csv	CSV artifact για audit ή περαιτέρω ανάλυση.
artifacts/diagnostics/regime_performance.csv	CSV artifact για audit ή περαιτέρω ανάλυση.
artifacts/diagnostics/residual_histogram.png	Plot artifact από το report/diagnostics.
artifacts/diagnostics/summary.json	Μηχανικά αναγνώσιμη σύνοψη metrics.
artifacts/diagnostics/threshold_grid.csv	CSV artifact για audit ή περαιτέρω ανάλυση.
artifacts/diagnostics/turnover_cost_timeseries.csv	CSV artifact για audit ή περαιτέρω ανάλυση.
artifacts/diagnostics/turnover_timeseries.png	Plot artifact από το report/diagnostics.

Artifact	Ρόλος
artifacts/diagnostics/turnover_vs_net_pnl.png	Plot artifact από το report/diagnostics.

Όλα τα logged runs αυτής της οικογένειας

Run	Strategy
ethusd_30m_lightgbm_garch_overlay_h24_structured_tail_alpha_v3_7_ehlers_trend_hybrid_20260705_163128_876008_ae3471e3	ethusd_30m_lightgbm_
ethusd_30m_lightgbm_h24_structured_tail_alpha_v3_7_ehlers_trend_hybrid_20260705_122927_309175_b6a5edd2	ethusd_30m_lightgbm_
ethusd_30m_lightgbm_h24_structured_tail_alpha_v3_6_ehlers_trend_replacement_20260705_122630_980027_bd9f9249	ethusd_30m_lightgbm_
ethusd_30m_lightgbm_h24_structured_tail_alpha_v3_1_remove_calm_abs_20260704_193715_173084_6a63d359	ethusd_30m_lightgbm_
ethusd_30m_xgboost_h24_structured_tail_alpha_v3_7_ehlers_trend_hybrid_20260705_162305_894106_c49abbed	ethusd_30m_xgboost_h
ethusd_30m_lightgbm_h24_structured_tail_alpha_v3_4_abs_range_bb_expansion_20260704_210256_965492_cd42de7e	ethusd_30m_lightgbm_
ethusd_30m_lightgbm_garch_vol_filter_h24_structured_tail_alpha_v3_7_ehlers_trend_hybrid_20260705_181216_114952_85d87588	ethusd_30m_lightgbm_
ethusd_30m_lightgbm_h24_structured_tail_alpha_v3_3_remove_calm_bb_expansion_20260704_194409_963904_67f68343	ethusd_30m_lightgbm_
ethusd_30m_lightgbm_h24_structured_tail_alpha_v3_2_remove_calm_rank_20260704_194247_008727_78d654d3	ethusd_30m_lightgbm_
ethusd_30m_lightgbm_h12_structured_tail_alpha_v3_7_ehlers_trend_hybrid_20260705_160827_328262_1bd6c114	ethusd_30m_lightgbm_
ethusd_30m_lightgbm_h24_structured_tail_alpha_v2_regime_gated_20260704_141159_183262_c78b2045	ethusd_30m_lightgbm_
ethusd_30m_lightgbm_h24_structured_tail_alpha_v3_5_gk_additive_20260705_111251_455043_5dd92491	ethusd_30m_lightgbm_
ethusd_30m_lightgbm_h24_structured_tail_alpha_v3_7_ehlers_trend_hybrid_tail_validation_20260705_130557_085537_646b742f	ethusd_30m_lightgbm_
ethusd_30m_lightgbm_h24_structured_tail_alpha_v3_7_ehlers_trend_hybrid_tail_validation_20260705_141427_384967_748825c0	ethusd_30m_lightgbm_
ethusd_30m_lightgbm_h24_structured_tail_alpha_v2_hysteresis_20260704_143416_406462_47f5016a	ethusd_30m_lightgbm_
ethusd_30m_lightgbm_h24_structured_tail_alpha_v3_7_ehlers_trend_hybrid_tail_validation_20260705_142225_670698_b9f447a5	ethusd_30m_lightgbm_
ethusd_30m_lightgbm_h24_structured_tail_alpha_v3_7_ehlers_trend_hybrid_tail_validation_20260705_142516_496102_80166fbd	ethusd_30m_lightgbm_
ethusd_30m_lightgbm_h24_trade_quality_meta_v1_20260704_143614_272498_421eebe2	ethusd_30m_lightgbm_
ethusd_30m_lightgbm_h24_structured_tail_alpha_v3_7_ehlers_trend_hybrid_tail_validation_20260705_142659_931684_6c1094e2	ethusd_30m_lightgbm_
ethusd_30m_lightgbm_h12_structured_tail_alpha_v3_7_ehlers_trend_hybrid_20260705_160212_543986_fd038ad8	ethusd_30m_lightgbm_
ethusd_30m_lightgbm_h24_structured_tail_alpha_v2_sparse_threshold_grid_20260704_140732_542073_e932ffe7	ethusd_30m_lightgbm_
ethusd_30m_lightgbm_h12_structured_tail_alpha_v3_7_ehlers_trend_hybrid_20260705_155945_510449_8dde6ff5	ethusd_30m_lightgbm_
ethusd_30m_lightgbm_h24_structured_tail_alpha_v3_7_ehlers_trend_hybrid_tail_validation_20260705_142758_740331_64c1477f	ethusd_30m_lightgbm_
ethusd_30m_lightgbm_h24_structured_tail_alpha_v2_no_raw_vol_levels_20260704_140951_265134_61cb2975	ethusd_30m_lightgbm_
ethusd_30m_lstm_h24_structured_tail_alpha_v3_7_ehlers_trend_hybrid_20260705_164947_169834_34a2a425	ethusd_30m_lstm_h24_

Πώς να αναπαραχθεί

Ενδεικτικά, κάθε config μπορεί να τρέξει με τον official runner:

```
docker compose run --rm app python -m src.experiments.runner
experiments/foundation_alpha/BEST_ethusd_30m_lightgbm_h24_structured_tail_alpha_v3_7_ehlers_trend_hybrid.yaml
```

Για serious comparison, κράτα σταθερά data snapshot, costs, split semantics, random seeds και config diff. Μην συγκρίνεις run που αλλάζει ταυτόχρονα target, features, model, signal threshold και backtest constraints χωρίς ablation table.